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Synthetic gene circuits and cellular decision-making in human pluripotent stem cells

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Abstract
Precise manipulation of human cell fate is a long-standing goal in biological engineering; it underpins efforts to advance new cellular therapeutics, and to develop disease models for fundamental research. Human pluripotent stem cells (hPSC) are emerging as the cells of choice for many of these applications. Currently, most advanced methods for manipulation of hPSC fate involve recapitulating developmental processes by mimicking stem cell niche-associated signals. Alternately, advances in engineering synthetic decision-making gene circuits provide an approach in which cell fate is directly controlled at the gene expression level. Here, we make the case that integrating these two engineering approaches represents an exciting opportunity to advance our fundamental understanding of cell fate control, and to accelerate new engineering strategies to manipulate cellular behavior for therapy.

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Introduction

During embryonic development, cellular decision-making yields remarkably ordered processes such as patterning, morphogenesis and organogenesis that ultimately transform a single zygote into a complex organism [1–3]. One of the key players in early development are pluripotent stem cells (PSC), transient cells that form at the blastocyst stage and give rise to the three germ layers that subsequently specialize into the various tissues forming the body [4]. Differentiation of PSC *in vivo* happens as discrete sequential cellular decision events with associated changes in gene expression patterns controlled by gene regulatory networks (GRN) [5,6]. The precision of these events is remarkable considering that the GRN respond to and integrate multiple noisy [7] and often conflicting cues. In 1998 the first report of successful *in vitro* culture of hPSC was published, demonstrating that human embryonic stem cells (hESC) could be isolated from the inner cell mass (ICM) of blastocyst-stage embryos and stably cultured in the presence of developmentally appropriate signals [8]. Interestingly, about 10 years later, the *de novo* generation of hESC-like cells from fibroblasts was reported [9]. These reprogrammed cells, called induced PSC (iPSC), were generated by introducing components of the hESC GRN into cell populations *in vitro*. The generation of hESC by mimicking the *in vivo* niche *in vitro*, and the generation of iPSC by artificially manipulating a differentiated cells' GRN, are two significant scientific achievements that lay the foundation for more advanced approaches to manipulate human stem cell fate and with this the idea to use stem cells as a renewable source of human cells to generate cellular therapeutics [10] and to model development and disease [11]. Certainly, unlocking this potential requires technologies for robust and scalable production of hPSC and hPSC-derived cell types *in vitro*. Thus, a long standing goal of stem cell bioengineering [12] has been to develop advanced strategies to mimic the developmental niche to manipulate cell fate by exposing cells to carefully controlled environments [10,13,14]. As applied to hPSC, this approach aims to produce cell types that are phenotypically and functionally similar to their *in vivo* equivalents by guiding cells through developmental decisions using exogenously provided and controlled signals; an approach defined here as "reverse engineering".

Interestingly, the concept of controlling cellular fate decisions also resonates with the field of synthetic biology. One of the often articulated aims of synthetic biology, is the generation of bespoke or "synthetic" cells augmented with novel functionalities by rationally engineering synthetic gene circuits using a well characterized chassis cell line as a foundation [15,16]. Synthetic gene circuits are small networks of interacting genes designed to perform a defined function inside single cells. To create these gene circuits, synthetic biology typically relies on well-characterized building blocks, bottom-up assembly and predictive modeling and prototyping, approaches here termed as "forward engineering".

Although stem cell bioengineering (by reverse engineering the developmental niche) and synthetic biology (by forward engineering synthetic gene circuits) use inherently different engineering approaches, both disciplines share the goal of predictably and efficiently controlling (human) cells in a proscriptive manner to enable new applications. In this review, we summarize some opportunities and limitations of these strategies and argue that significant synergies may emerge from their integrated use.

Stem cell bioengineering: reverse engineering of developmental processes

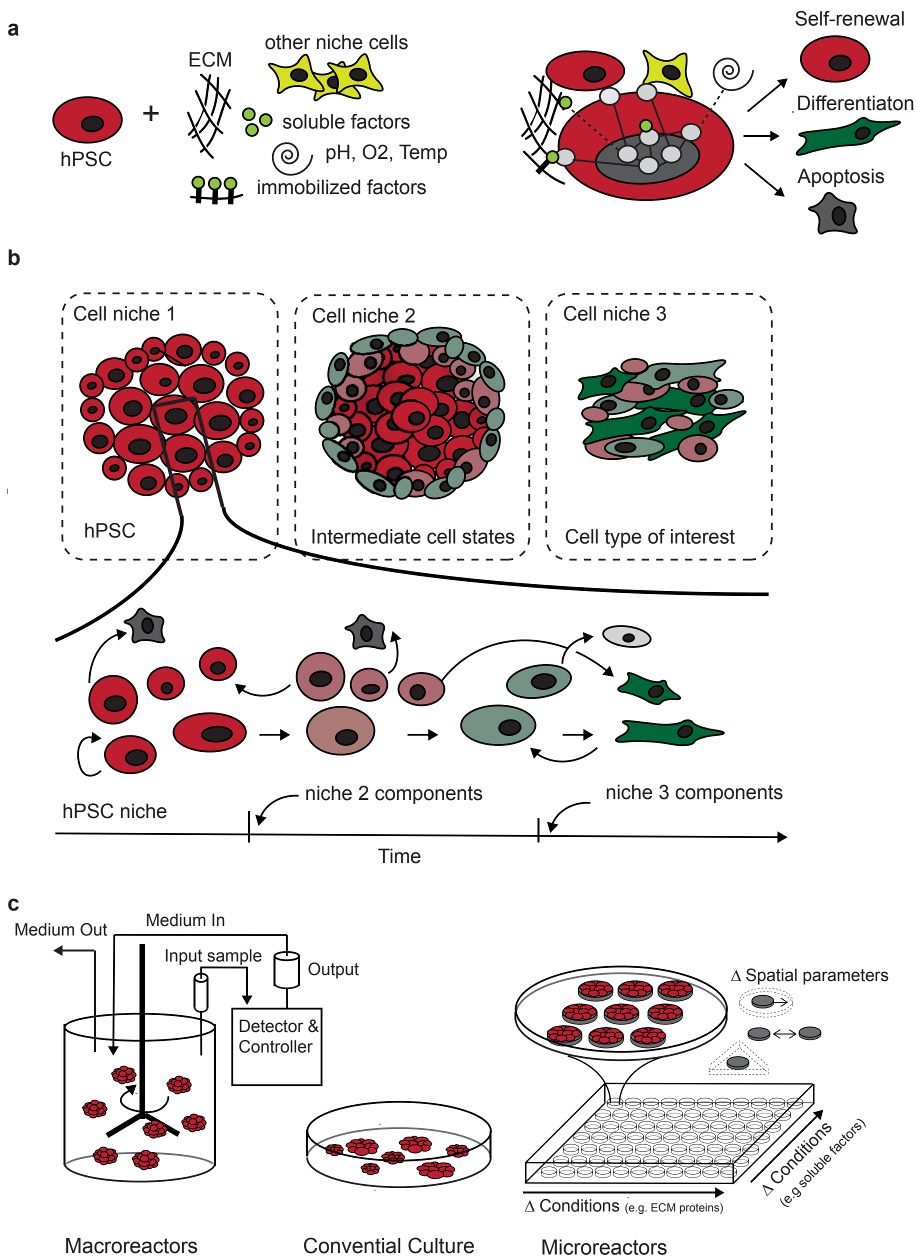
Typically, stem cell bioengineering aims to mimic, or "reverse engineer", the *in vivo* developmental niche to guide cells toward functional specialized cell types [13] and more recently toward tissues [17] and organoids [18]. Engineering the hPSC niche involves the coculture with components such as ECM, other niche cells (or feeder cells), soluble factors (such as growth factors or cytokines) and immobilized factors (such as ligands or protein components of the ECM). Additionally, physicochemical factors that influence the cell state such as CO_2, pH and temperature are tightly controlled (Figure 1a, left) [14,19]. The biological components can be of natural or synthetic nature and are mobilized in hopes of simulating signaling events occurring during development. This means the niche components have to mimic numerous cell-cell-, cell-ECM-, and biochemical (paracrine- and autocrine signaling) interactions. These interactions usually directly influence the GRN and the associated decision-making tasks (Figure 1a, right).

Usually, detailed hypotheses of signaling pathways activated in mice or other model systems are translated to the human system [20]. Early cell patterning events in the embryo are commonly recapitulated *in vitro* by aggregating hPSC to form 3D stem cell aggregates sometimes called embryoid bodies. Alternatively, hPSC are coaxed toward specific cell types using feeder cells and more recently, serum free media and hydrogels that imitate the fate-inducing activity of the cellular niche [21]. Administration of small molecules, growth factors or cytokines is also commonly used to manipulate the underlying signaling pathways that support cell transitions toward desired lineages [13].

Progress of the field is reflected in more widespread adoption of bioprocessing solutions, introduction of quality and safety standards, and several stem cell-derived products in clinical trials [10,22,23]. Yet, well established protocols for hPSC-derived cell products are limited to a handful of therapeutically relevant cell types including multiple neuronal cell types, cardiac and pancreatic-islet cells [13]. Yet in many cases these protocols are challenging to scale in a cost-effective manner, and/or they require an *in vivo* maturation step to be functional. Further, some cell types require maturation or induction via 3D reconstitution of the corresponding tissue (e.g. mature cardiac cells [24]), or are too rare to be detected, characterized or maintained *in vitro* (e.g. hematopoietic stem cells HSC [25]). Additionally, most protocols apply step-wise external stimulation to dynamically changing and often unsynchronized cells, irrespective of the current cell state (the GRN) of individual cells (Figure 1b). Indeed, it is likely that this is one of the reasons why protocols often have low yields, rely on subpopulation selection, or support heterogeneous cell types, despite the input cells being exposed to apparently similar external stimuli.

These challenges have recently led to efforts and technologies to better characterize and simulate the dynamic microenvironment and the associated GRN state during *in vitro* propagation and differentiation. Example technologies include bioreactor systems for robust production of hPSC [26,27] and hPSC-derived cell types such as cardiomyocytes [28] (Figure 1c, left). Other technologies are based on microscale reaction systems to rapidly and economically explore complex environmental parameters and precise spatiotemporal control of the niche. Such techniques involve micropatterning [29–31], microfluidics [32] and synthetic cell–cell interaction arrays [33]. Micropatterning, for example, is a powerful technology to control colony size, shape and separation that enables the manipulation of living cells in a defined spatial context over time in a high-throughput (HT) fashion (Figure 1c, right). A recent study, for example, used micropatterning to significantly improve the production of hPSC-derived hematopoietic progenitor cells by systematic manipulation of colony size and separation [29]. Subsequently, the platform was used to screen various compounds and led to the identification of a chemokine (IP-10) that inhibits differentiation of hematopoietic progenitor cells in unpatterned cultures but not in patterned colonies of defined sizes, thus giving insight in molecular mechanisms responsible in the spatial dependence of hematopoietic progenitor

Figure 1



Reverse engineering the developmental niche. **a)** hPSC are cultured with defined niche components (left). Niche components interacting with hPSCs' GRN to change its fate (right). **b)** Differentiation of hPSC to a desired cell type occurs through multiple decision-making events. Top: colony perspective of a hypothetical two-step differentiation protocol. Bottom: Single cell view and possible fates of individual cells. Usually, a heterogeneous hPSC population moves toward a heterogeneous end population, were only a few cells represent the cell type of interest (dark green). The differentiation protocol usually provides a discrete change in the cellular niche, while individual cells are dynamically changing. **c)** Technologies for increased hPSC cell fate control. Conditional culture (middle) in comparison with more advance bioreactors (left) and micropatterning technique (right).

production [29]. Aside from the ability to rapidly screen for spatial parameters and compounds that might affect the production of hPSC-derived cell types, such micropatterning techniques are useful to study basic cellular and molecular mechanisms governing human cell fate transitions. For example, two recent studies describe predictive computational models of an extra-cellular cytokine reaction-diffusion network and connect these components to the gradient formation in micropatterned colonies that mimic early gastrulation events using hPSC [30,31].

Indeed, an increased understanding of the decision-making trajectories of hPSC during differentiation is critical in developing robust differentiation protocols. Successful implementation of such approaches will rely on single cell 'omics technologies and advanced statistical and machine learning strategies to identify key nodes governing cell transitions. Recent efforts in the field have combined predictive computational models with single cell data to elaborate the core GRN network during cell state transitions [34,35]. Recently, this approach has been expanded to connect the core GRN with the essential cues of the extra-cellular environment to predict if, and under what conditions, a cell state transition can be encouraged [36]. Current efforts and remaining challenge lies in extending these single-cell simulations to cell populations that include parameters such as cell density, cell number and distance between cells or colonies (so called multi-paradigm modeling approaches) [37]. The major challenge, however, will be to validate and translate this knowledge into technologies that allow to produce hPSC-derived cell types in a scalable and predictive manner.

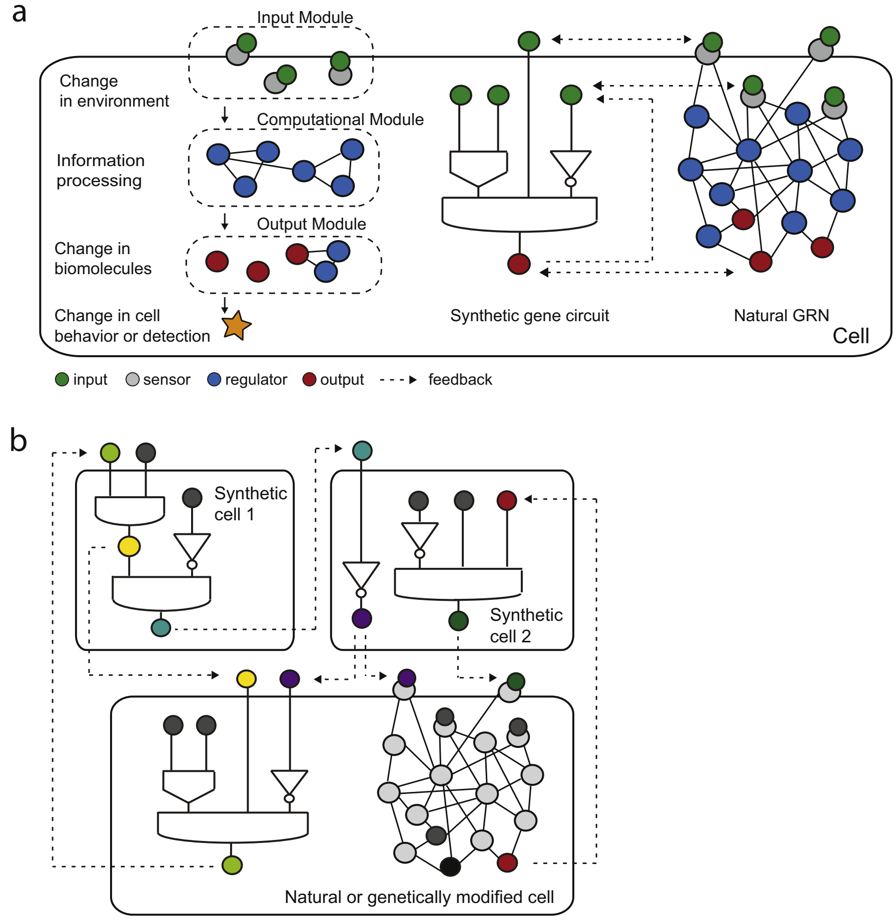
Synthetic biology: forward engineering of genetic decision-making gene circuits

Although the term synthetic biology is sometimes associated with "genetic engineering" or "genome editing", the origin, approach and vision of this field is far broader. Here we focus on decision-making synthetic gene circuits as an opportunity where synthetic biology can impact and integrate with stem cell bioengineering. Indeed although the idea of decision-making molecular systems was first introduced in the context of *in vitro* molecular computing [38,39], significant efforts are underway to combine these molecular systems with human cells including stem cells [40]. The goal of this approach is not to reverse engineer the natural decision-making gene network (GRN) but to implement artificial decision-making circuits that co-exists with, or replaces, the natural GRN (Figure 2a). Such synthetic gene circuits, as all responsive biological and artificial systems, can be broadly subdivided into input, computational and output modules (Figure 2a, left) [40,41]. The input module is composed of sensor components that can include a DNA response element recognizing a transcription factor, a miRNA target sequence on the mRNA, or the binding domain of a protein recognizing small molecules. The computational module is composed of various well-defined regulatory components that transmit and process the signal(s) and decide whether one or multiple output molecule/s will be produced (or repressed), and at what dosage. The change in the output subsequently leads to a change in cell behavior or in the environment, or reports on the input molecules. The relationship between the input(s) and the output(s) in a circuit is typically approximated by a mathematical function that is specified before the circuit is built, according to a specific task the circuit is designed to implement. This function is usually referred to as a "computation" performed by the circuit, and it is understood that the actual relationship between the inputs and outputs as measured in an experiment can be much more complex than this predefined function. These "computations" are often formulated as Boolean logic functions to approximate input—output relationships. One reason for using Boolean functions as high-level specification of circuit computations, is the fact that biochemical interactions are often well described by Hill functions, which are themselves well approximated by Boolean functions [42]. This strategy provides a convenient solution for experimental characterization and validation of complex input—output relationships, thus allowing to develop gene circuits that integrate multiple inputs cues (Figure 2a middle). However, many other types of computations, including analog ones, can be implemented using synthetic gene circuits [43]. Even with the understanding that the pre-specified computation is only an approximation, constructing a circuit whose experimental input—output relationship is well-approximated by this a priori specification remains a challenge.

Certainly, one of the goals of synthetic biology is to integrate gene circuits that perform an abstract computational function into the cellular context to build autonomous decision-making gene circuits. Such circuits would respond to physiological input cues (coming from the cell environment, extra-cellular matrix (ECM), or endogenously expressed factors) and as a result, produce physiological output molecules that would lead to a change in cellular behavior or environment, affecting neighboring cells and/or the niche. Importantly, input and output cues will continuously feedback to the endogenous GRN, while the synthetic framework (ideally) does not interfere with the cellular context (Figure 2a, right). Extending this single cell view to multicellular systems composed of engineered cells that communicate with each other through pre-defined or synthetic molecules can in theory be employed to encourage (pre-specified) multicellular organization (Figure 2b). This would lead to a controlled "cell collective" that, if programmed correctly, can respond appropriately to limited external cell niche components (reviewed in Ref. [44]).

While almost limitless possibilities can be imagined using such engineered systems, implementation of these systems in human cells is still in its infancy. Indeed, the current focus on synthetic gene circuits relies on achieving accurate operation in unconnected single cells, using HEK or CHO as the workhorse to demonstrate proof of concept performance. The

Figure 2



Synthetic cellular decision-making systems. **a)** Schematic of cellular decision-making gene circuits with different levels of abstraction inside a cell. Left: modular design of genetic decision-making gene circuits. Middle: genetic decision-making circuit illustrated as logic gate. Constant interplay and feedback with the endogenous GRN (right) is indicated. **b)** Multi-cellular decision-making. Cells are dynamically influencing each other and the extra-cellular environment. Arrow: Output of sender cell has a negative or positive impact on output production of receiver cell.

majority of gene circuits that have been implemented in human cells today are simple switches that transmit single input cues using RNAi-, ribozyme-, recombinase-, transcription factors, or CRISPR/Cas9 based components or alternatively couple a natural signaling pathway to a synthetic inducer or vice versa (reviewed in Ref. [45]). These switches can be directly employed to control a desired gene (mostly by external administration of synthetic small molecule) or used as a building block to build more complex gene circuits. For example, several more complex systems have been implemented to integrate multiple inputs (up to 6 inputs have been achieved) in logic functions using the above introduced design principles [46–49]. Other, recently developed synthetic frameworks have allowed us to produce multi-input-multi-output gene circuits [50,51]. Such frameworks allow, in principle, the implementation of fairly complex decision-making tasks inside individual cells. For example, the "Bow-tie" framework allows integration of multiple endogenous microRNAs in an AND-logic function and then, in response to these microRNAs, controls multiple proteins dynamically. Thus the expression of a desired output molecule can be restricted to a cell-type specific microRNA combination, either to detect or remember this cellular state or to conditionally control physiological output cues [50]. Another interesting strategy is a framework called "BLADE" (Boolean logic and arithmetic through DNA excision) that allows construction of more than 100 distinct logic circuits. One of several topologies proposed in this study allows the control of four distinct gRNAs in response to different combinations of two small molecule inputs allowing for the precise activation of four endogenous genes via dCas9-VPR [51]. Excitingly, efforts to create gene circuits that extend beyond single cells have recently led to synthetic cell–cell communication systems [52,53] and cell–cell contact signaling pathways [54] (discussed in Ref. [55]).

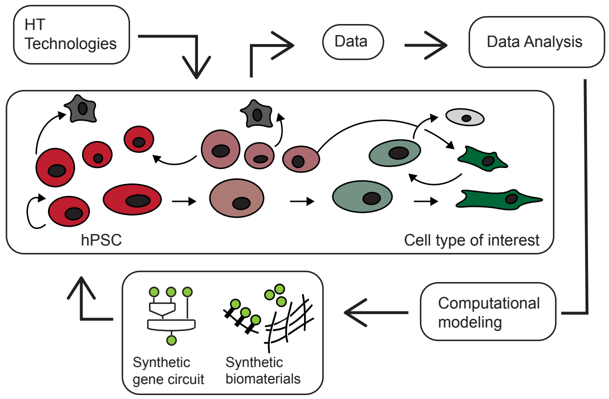
Despite growing numbers of successfully implemented gene circuits in human cells, it has proven difficult to translate proof-of-concept implementations, especially for the more complex gene circuits developed in HEK or CHO, to other human cell types. Indeed, there are only a handful of synthetic gene circuits that are integrated into cellular contexts to produce biologically meaningful decision-making tasks. Examples include the conditional killing of cancer cells [47], control of T cell proliferation [53,56], and so called prosthetic circuits that are designed to restore homeostasis of metabolites, cytokines and growth factors [57]. Last but not least, there are several synthetic systems designed for and integrated in hPSC (explained in more detail below). One of the reasons implementation and adoption to different human cell types is difficult (in addition to various technical limitations [16]) is that most genetic components, especially those that rely on transcriptional regulation, perform very differently in different cell types because they are not sufficiently well isolated from the endogenous GRN. If it is required that gene circuits are stably integrated in the genome, as would be the case for a stem cell differentiation protocol, chromosomal interference and epigenetic silencing complicates the production of robustly performing circuits [58]. Another challenge and uncertainty in using orthogonal non-human components is immune response of the host. Therefore, it is currently very challenging to generate faithful gene circuits for therapy or for applications in a dynamic biological context (when cells are undergoing state transitions). Exciting work, such as the development of human artificial chromosomes (HAC) [59], orthogonal transcriptional machinery [60], network motives that reduce noise [61], strategies for module insulation [62], and/or the use of a well understood chassis cell line [16], might offer solutions for such problems. However, eventually genetic circuits that are designed to increase our understanding of cellular mechanisms, or to recognize or produce physiological signals, will have to be engineered in the cell type where its function is intended. Progress overcoming these has been hindered, in part because many human cell types and developmental niches are not accessible for testing, and cells of interest typically cannot be propagated for a sufficiently long time to establish and test new gene circuits. Excitingly, hPSC can offer a solution to this problem by providing human cell models (including various cell types, tissues and organoids) for testing and improving synthetic tools and gene circuits.

How synthetic decision-making gene circuits can advance stem cell engineering

The stem cell field has a long history in genetic engineering using mouse PSC and today various genetic tools and approaches are being translated to hPSC. Traditionally, gene knock-outs, fluorescent labeling of endogenous genes or sensing of transcription factors or signaling pathways have been used to study essential genes that maintain the pluripotent state or lead to cell state transitions [63]. Another focus, that has recently received new attention due to the introduction of CRISPR/Cas9-based approaches [64,65], lies in developing genetic lineage tracking systems to track lineage choices in single live cells [66]. Other genetic methods have been developed to overcome challenges in producing hPSC-derived cells, most common being the defined control of key-transcription factors to manipulate hPSC transitions to differentiated cells. Usually, Dox inducible genetic switches controlling ectopic transcription factors (TF) [67–69] or Cas9-mediated activation of endogenous TF [70–72] are used. A recent study, for example, developed a platform for engineering hPSC lines for stable Dox-inducible expression of key TFs from two safe harbor sites in the human genome [73]. Providing standardized hPSC acceptor lines allows us to engineer various hPSC lines with identical genetic backgrounds and more predictive control over the synthetic TF. Indeed, with this approach, it has been demonstrated that controlled overexpression of lineage specific TFs can increase the efficiency of hPSC differentiation to neurons, myocytes and oligodendrocytes [73]. Another exciting study reverse engineered the temporal and dose specific interplay of Pdx1, Ngn3 and MafA (based on knowledge from mouse studies) to increase the hPSC-derived production of human beta-like cells [74]. Finally, another key interest in the stem cell field is to produce clinical-grade hPSC using genetically engineered fail-safe switches that ablate hPSC but not differentiated cells upon transplantation [75,76]. For example, one recently established switch puts a resistance gene under the control of an endogenous miRNA (miR-302) that is expressed only in hPSC but not in differentiated cells [76].

These examples are a good representation of the benefits of genetic tools to control, report on - or respond to endogenous cellular decision-making processes to address unanswered questions in developmental biology, to increase hPSC-derived cell production or to develop cell therapeutics with genetically engineered features. Yet, autonomous synthetic decision-making systems remain to be integrated in these approaches. Indeed such systems are likely able to guide individual cells dynamically toward the desired lineage [77] (Figure 3). For example, in the production of mature cell types, PSC undergo multiple cell state transitions that are guided by multiple key regulatory genes in a temporal fashion. These key regulators could serve as input cues for sequentially connected decision-making circuits. For example, a possible function of a gene circuit would be, if TF1 AND TF2 are expressed then induce TF3, if TF3 AND NOT TF2 AND NOT TF1 is expressed induce TF4 etc. [78]. These functions can become more complex over time as the knowledge of

Figure 3



Integrating synthetic decision-making gene circuits in hPSC derived reverse engineering approaches. High-throughput (HT) technologies can be used to generate data that capture dynamic interplay of intracellular and extracellular molecular species. Knowledge increases in cycles of data interpretation, production of predictive models and replacement of natural with synthetic components.

cell state transitions becomes more accurate and the ability to produce larger and more robust gene circuits increases. Eventually, such systems, if they indeed lead to increased production of the desired cell type, validate assumptions of developmental trajectories and mechanisms of cell state transitions. The same approach is feasible for the generation of tissues and organoids but it requires an increased level of complexity that includes cell—cell, cell-ECM interactions and precise spatio-temporal control of the signaling cues [44].

Another application of an autonomous decision-making gene circuit would be to first capture and then lock a cell in a desired state. This could be a state within a heterogeneous hPSC population [79] or a desired hPSC-derived cell type during differentiation. For example, it may be desired to lock hPSC in a homogenous naïve-like state [79] — a state that may be favored over the primed state in applications requiring increased growth rates and single cell survival frequencies.

Eventually the engineering of decision-making gene circuits should become an integral part of iterative cycles in reverse engineering the hPSC niche for differentiation or for stabilization of defined hPSC states during cultivation (Figure 3). These iterative cycles involve: 1. Generation of data during cell state transition. Ideally this should be unbiased HT single cell data measuring both the components of the core GRN and the cellular niche. 2. Analysis of data and identification of core components by statistical and machine learning approaches. This information then feeds into predictive multi-paradigm computational models to predict and simulate the conditions associated with a given cell state transition. 3. Validation of the computationally predicted components and proposed mechanism for cell state transition by synthetic engineered systems. These systems would involve both synthetic gene circuits augmenting the core GRN and synthetic biomaterials replacing defined extracellular or niche components (Figure 1b). Importantly, once synthetic systems are implemented, the overall differentiation or cultivation process can be more easily perturbed by varying the essential synthetic components toward the trajectory that is known to lead to rapid and robust transition (in the case of differentiation) and or to lock the cells in a given state (in the case of cultivation in defined states). This information can feedback directly into the predictive computational model. In repetitive cycles of this approach, assuming data of adequate quality, we gain a better understanding of how cell fate decisions are made *in vitro* and can then use this knowledge to generate protocols and synthetic systems that guide the cells dynamically and robustly through changing environment.

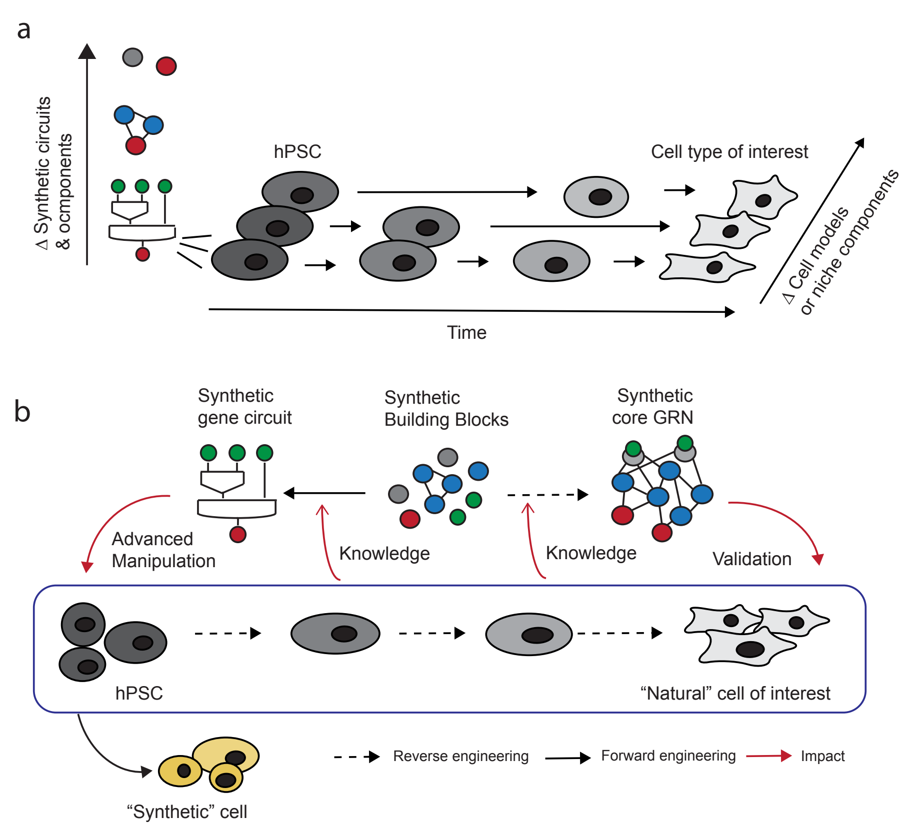
How stem cell engineering can advance the development of synthetic gene circuits in human cells

This and several other excellent reviews [10,18,44,55,77] have outlined the potential benefits of rationally engineered gene circuits to the stem cell field. However, this relies in the first place on our ability to rapidly implement gene circuits in human cells and on the circuits' ability to perform robustly even if the cell undergoes state transitions or is being exposed to new extracellular cues or niches. This is, at the moment, quite challenging, as outlined above. However, we hypothesize that moving synthetic circuit engineering efforts to hPSC and using established cellular platforms and technologies from the stem cell field may help overcome many of the limitations encountered in generation of gene circuits in human cells.

The advantages are twofold. First, the self-renewal capacity of hPSC allows us to engineer complex gene circuits and optimize synthetic components to the level comparable to the current CHO and HEK workhorses, with the added advantage that hPSC typically possess normal karyotypes. Additionally, ongoing efforts in improving, standardizing and simplifying hPSC culture conditions and the ability to keep hPSC in desired and well defined cell states, make hPSC a more standardized cell source for implementation of gene circuits. Indeed, with the iPSC technologies, standard sharing and safety policies are in place for various iPSC lines, allowing hPSC to enter most typical non-stem cell labs. Second and most importantly, stem cell bioengineering field provides protocols to differentiate hPSC to various cell-types, tissues and organoids. These cell types and cellular model systems offer an unprecedented platform for testing and improving synthetic components and gene circuits in cell types that cannot be easily accessed or manipulated. Thus, one can imagine stable integration of a whole library of synthetic components and gene circuits in hPSC (ideally in well-defined harbor sites or on a HAC) and then differentiating these hPSC lines to generate tissues and organoids. This would allow us to study the genetic components in various cell types and in an integrated 2D or 3D context. This in turn can give information on how tissue-specific internal cues, or neighboring cells or components of the ECM, would affect the synthetic system. Thus, hPSC-derived cellular platforms could help engineer synthetic parts to be more robust (i.e. functionally identical across many tissues) or to function tailored in a desired human cell type. Indeed, synthetic biology has not extensively explored this opportunity yet, partly due to technical constraints and laborious protocols for hPSC differentiation and the low efficiency of generating stable hPSC lines. With advanced tools for genome editing such as CRISPR/Cas9 and strategies to deliver and stably integrate synthetic component (using hPSC line with characterized landing pads or HAC) these hurdles are becoming lower in future.

Taken together, the engineering of complex decision-making gene circuits is directly linked to the ability to produce hPSC derived cell types and is an integral part of the iterative cycles of hPSC-derived cell production (Figure 4). First, synthetic biology tools help to increase our knowledge of mechanisms guiding human cell fate choices. Second, synthetic building blocks can be engineered to function properly in a given cell type or when cells underdo cell transitions. Combining the knowledge of both should allow us to reverse engineer complex “natural” core GRN. This in turn can validate presumptions of natural core GRN (the kind, number and interconnectivity) during state transition more precisely than abstract (logic) gene circuits (as introduced above). Ultimately, being able to produce robust gene circuits-either abstract or reverse engineered ones-will allow us to produce “synthetic” cells (using hPSC or another convenient cell as a source) in a forward engineering approach (Figure 4). This approach can focus on performing a predefined cellular function rather than mimicking the natural phenotype (as is the aim today). Importantly, such synthetic cells can be engineered for tailored operation in an *in vivo* context using available

Figure 4



Integrating hPSC derived cellular platforms in synthetic gene circuit engineering. **a)** Use of hPSC and hPSC differentiation protocol to test synthetic building blocks or synthetic circuits. **b)** Cycle of dependence for forward engineered synthetic circuits on reverse engineered hPSC derived cell types. Better synthetic building blocks and better understanding of human cell fate control can lead to synthetic gene circuits that more accurately represent “natural” GRN and to the ability to engineer “synthetic” cells with predefined function in a more rapid forward engineering approach.

hPSC derived cell, organoid and tissue models. Therefore this opens new opportunities for generating cellular therapeutics that have synthetically integrated mechanisms for studying or regenerating disordered cellular functions [10] and at the same time could contain safety mechanisms to remove cells that become tumorigenic or do not function properly *in vivo* [75,76].

Conclusions

Synthetic biology and stem cell bioengineering are poised to overcome challenges that arise as a consequence of our limited understanding of the dynamics and molecular mechanisms governing human cell fate decisions. We conclude that the two engineering approaches naturally complement each other and their mutual cross-pollination can increase our knowledge of cellular decision-making in human cells. We hypothesize that, as a result, synthetic gene circuits will become more accurate, complex, and able to reverse engineer human GRN that govern cell state transitions. This in turn can lead to rapid and precise forward engineering of human cell fate and eventually allow us to advance to a new stage of stem cell bioengineering — one where both the cell and its niche can be prospectively designed to code cell fate not only *in vitro* but eventually *in vivo* to restore aberrant cellular functions and regenerate tissues.

Conflict of interest

The authors declare no competing financial interest.

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- •• of outstanding interest

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